

IILM UNIVERSITY, GREATER NOIDA, U.P.

School of Computer Science and Engineering

REPORT OF

**ARTIFICIAL INTELLIGENCE INTERNSHIP:
SENTIMENT ANALYSIS OF CUSTOMER REVIEWS USING MACHINE
LEARNING AND NLP**

Submitted To

**School of Computer Science and Engineering
IILM University, Greater Noida, Uttar Pradesh**

In partial fulfilment of the requirement of the degree of
Bachelor of Technology in Computer Science and Engineering

Name of Student	:	DISHA CHAUDHARY
Roll Number	:	25SCS1003001639
Programme / Branch	:	B.Tech. (CSE)
Batch / Session	:	2026–27
Host Organisation	:	Codec Technologies Pvt. Ltd.
Designation	:	Artificial Intelligence Intern
Duration	:	08/08/2026 to 08/09/2026 (One Month)

2026–27

**AICTE & ICAC Approved Internship | National Internship
Portal**

CANDIDATE'S DECLARATION

I, **DISHA CHAUDHARY** (Roll No.: **25SCS1003001639**), do hereby declare that the internship report titled "**Artificial Intelligence Internship: Sentiment Analysis of Customer Reviews using Machine Learning and NLP**" has been completed by me to fulfil the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** from the School of Computer Science and Engineering, IILM University, Greater Noida, Uttar Pradesh.

The work recorded in this report is based on the one-month Artificial Intelligence internship undertaken at **Codec Technologies Pvt. Ltd.** from **08 August 2026 to 08 September 2026** under an AICTE and ICAC approved programme listed on the National Internship Portal. All the references have been quoted, and I have not taken as such any material from any other source without acknowledgement.

I affirm that this report is my own and has not been submitted to any other institute for any degree or diploma requirement. Further, I shall be solely responsible for any kind of copyright violation in this regard.

Signature of the Student

Student Name: DISHA CHAUDHARY

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Programme: B.Tech. (CSE)

Date: 10/09/2026

Place: Greater Noida

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I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Programme. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my sincere thanks to **Codec Technologies Pvt. Ltd.** for providing a structured, AICTE and ICAC approved Artificial Intelligence internship, and to **Dr. Anurag Shrivastava**, Programme Manager, for overall academic and professional guidance associated with the programme. I also thank the mentors and content team of Codec Technologies for laboratory-oriented exposure to Python, machine learning, deep learning and natural language processing.

I am grateful to the **School of Computer Science and Engineering, IILM University, Greater Noida**, for permitting me to undertake this internship in partial fulfilment of the B.Tech. (CSE) degree and for the academic environment that made this documentation possible.

I express my gratitude to my parents for their blessings, patience and constant encouragement.

Any merit in this report belongs to those who taught and guided me; the remaining shortcomings are mine alone.

Signature: _____

Student Name: DISHA CHAUDHARY

Roll No.: 25SCS1003001639

INTERNSHIP COMPLETION CERTIFICATE

A true copy of the Certificate of Internship issued by Codec Technologies Pvt. Ltd. is reproduced below. As specified in the report format, this industrial certificate page does not carry a printed page number.

Particulars as recorded on the certificate: Awarded to **DISHA CHAUDHARY**; organisation **Codec Technologies Pvt. Ltd.**; designation **Artificial Intelligence Intern**; duration **08/08/2026 to 08/09/2026**; programme **one-month AICTE and ICAC approved internship**; Programme Manager **Dr. Anurag Shrivastava**; AICTE ID **CORPORATE6759d549ce50e1733940553**; Google for Education Partner; National Internship Portal.

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CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

Artificial Intelligence (AI) is the body of computing methods that enable machines to perceive data, infer patterns and support decisions that would otherwise require human judgement. Within AI, Machine Learning (ML) learns parameters from examples, while Deep Learning (DL) uses layered neural networks to represent images, language and other high-dimensional signals. The one-month internship completed by candidate **DISHA CHAUDHARY** (Roll No.: **25SCS1003001639**) at **Codec Technologies Pvt. Ltd.** was designed around this progression: from Python-based data handling, through classical supervised learning and model evaluation, to neural networks, convolutional models, natural language processing (NLP) and a short introduction to reinforcement learning.

The internship was undertaken from **08 August 2026 to 08 September 2026** as an Artificial Intelligence Intern under an AICTE and ICAC approved scheme visible on the National Internship Portal. The academic purpose of the present report is to convert that structured industrial exposure into a coherent project narrative suitable for assessment by the School of Computer Science and Engineering, IILM University, Greater Noida.

Customer-generated text—product reviews, service feedback and social comments—is now a primary operational data source for organisations. Manual reading does not scale, and unsystematic sampling introduces bias. Sentiment analysis, a core NLP task, assigns polarity (commonly positive, negative or neutral) to an utterance so that product, support and marketing teams can monitor opinion in near real time. The capstone chosen for this internship therefore integrates the month's tools into a single, auditable pipeline: ingest labelled reviews, clean and vectorise text, train supervised classifiers, evaluate them with standard metrics, and document limitations. Parallel notebooks in regression, clustering and image classification were retained as evidence of syllabus coverage, but the unifying project theme remains sentiment analysis of customer reviews.

4.2 Organisation Profile

Codec Technologies Pvt. Ltd. is an information-technology training and internship provider that operates skill programmes in software, data and artificial intelligence, including internships aligned with the All India Council for Technical Education (AICTE) and listed through the National Internship Portal. The organisation describes itself as a Google for Education Partner and issues offer letters, completion certificates, training certificates and related academic artefacts on successful completion of an enrolled internship track.

Public programme literature for the one-month Artificial Intelligence internship specifies a four-week curriculum: (i) foundations of AI and Python for data handling; (ii) machine-learning basics with scikit-learn and evaluation; (iii) neural networks, TensorFlow/Keras and convolutional networks for images; and (iv) NLP, a brief reinforcement-learning module, and a capstone such as an image classifier, a chatbot or a sentiment-analysis model. Headquarters are published at 211B, Saki Vihar Road, West Chandivali IT Hub, Powai, Mumbai, Maharashtra 400076, with additional regional offices. The internship completion certificate issued to the candidate is signed by Dr. Anurag Shrivastava, Programme Manager, and carries AICTE corporate identifier CORPORATE6759d549ce50e1733940553.

For a B.Tech. (CSE) student, the value of this host setting is not a large-scale production deployment inside a product company, but a time-boxed, syllabus-mapped laboratory in which each major family of AI methods can be

implemented, measured and written up. That constraint is treated honestly in this report: the work is educational and capstone-oriented, yet it follows industrial toolchains (Python, scikit-learn, TensorFlow, NLP libraries) that transfer to subsequent academic projects and internships.

4.3 Problem Statement

Organisations accumulate large volumes of unstructured English reviews. Without an automated method, it is difficult to (a) quantify overall sentiment, (b) detect sudden negative shifts, and (c) compare models fairly. At the same time, a one-month intern cannot build a full multilingual production stack. The problem addressed in this internship is therefore stated as follows:

Design, implement and evaluate a supervised machine-learning pipeline that classifies customer-review text into sentiment classes, after systematic preprocessing and feature extraction, while also completing the prescribed AI internship modules in data handling, classical ML, neural networks, CNN, NLP and introductory reinforcement learning.

The problem has three practical difficulties. First, raw text is noisy (HTML fragments, slang, punctuation and class imbalance). Second, bag-of-words and TF-IDF representations ignore word order, so a neural baseline is needed for comparison. Third, accuracy alone is misleading when classes are skewed; precision, recall, F1-score and the confusion matrix must be reported together. The internship treats these difficulties as first-class engineering tasks rather than as afterthoughts.

4.4 Project Objectives

The internship and the capstone were guided by the following objectives:

- To study the relationship among Artificial Intelligence, Machine Learning and Deep Learning, and to apply Python with NumPy and Pandas to realistic tabular and text-oriented datasets.
- To implement supervised learners (linear regression for continuous targets and logistic regression for categorical targets) using scikit-learn, and to practise train-test splitting, cross-validation and hyperparameter awareness.
- To construct and train a simple neural network in TensorFlow/Keras and a baseline Convolutional Neural Network for image classification, thereby covering the spatial-learning module of the approved syllabus.
- To apply NLP preprocessing (tokenisation, normalisation, stop-word handling and vectorisation) and to train a sentiment classifier on customer-review text as the unifying capstone.
- To evaluate models with accuracy, precision, recall and F1-score, and to document architecture, limitations and ethical cautions in a form suitable for university assessment.
- To complete all formalities of the one-month AICTE and ICAC approved internship, including the industrial certificate attached to this report.

4.5 Scope of the Project

In scope: The work covers English-language review texts, binary or three-way polarity labels, classical ML baselines, a neural comparison model, standard metric reporting, and parallel syllabus notebooks (regression, a small CNN, and a tutorial Q-learning grid-world). Data handling is limited to publicly available or programme-provided CSV corpora and synthetic illustrative splits suitable for academic demonstration.

Out of scope: The internship does not include deployment as a public web service, real-time streaming, multilingual or code-mixed Indian-language models, transformer fine-tuning at production scale, or access to Codec Technologies' internal customer databases. Reinforcement learning is treated as a conceptual laboratory, not as the scoring engine of the sentiment product. These exclusions keep the one-month timeline feasible and academically honest.

Assumptions: Labels in the review corpus are taken as ground truth; hardware is a personal computer or hosted notebook runtime; and evaluation is offline (batch), not A/B tested on live users.

4.6 Technologies and Tools Used

Tool selection followed the official one-month Artificial Intelligence internship syllabus of the host organisation so that every library used in the capstone had already been introduced in weekly modules.

S. No.	Technology / Tool	Purpose in the Internship
1	Python 3	Primary language for scripts, notebooks and model code
2	NumPy and Pandas	Array computation, tabular wrangling and missing-value handling
3	Matplotlib and Seaborn	Exploratory visualisation of distributions, correlations and metrics
4	Scikit-learn	Regression, classification, clustering, pipelines and cross-validation
5	TensorFlow and Keras	Feed-forward networks and a baseline convolutional image classifier
6	NLTK / spaCy	Tokenisation, normalisation and linguistic pre-processing for reviews
7	Jupyter / Google Colab	Interactive experimentation, GPU-backed training and documentation
8	Git (basic) & CSV datasets	Versioning of notebooks and storage of labelled review corpora

Table 4.2: Technologies and tools used during the internship

4.7 System Architecture

The capstone is organised as a layered pipeline rather than as a single script. The data layer stores raw CSV reviews, cleaned frames and serialised vectorisers. The learning and NLP layers transform strings into numerical features and fit estimators. The application layer encodes internship workflows training, comparison and reporting. The presentation layer exposes plots and predicted labels inside notebooks. This separation made it possible to replace a logistic-regression head with a small neural network without rewriting ingestion code.

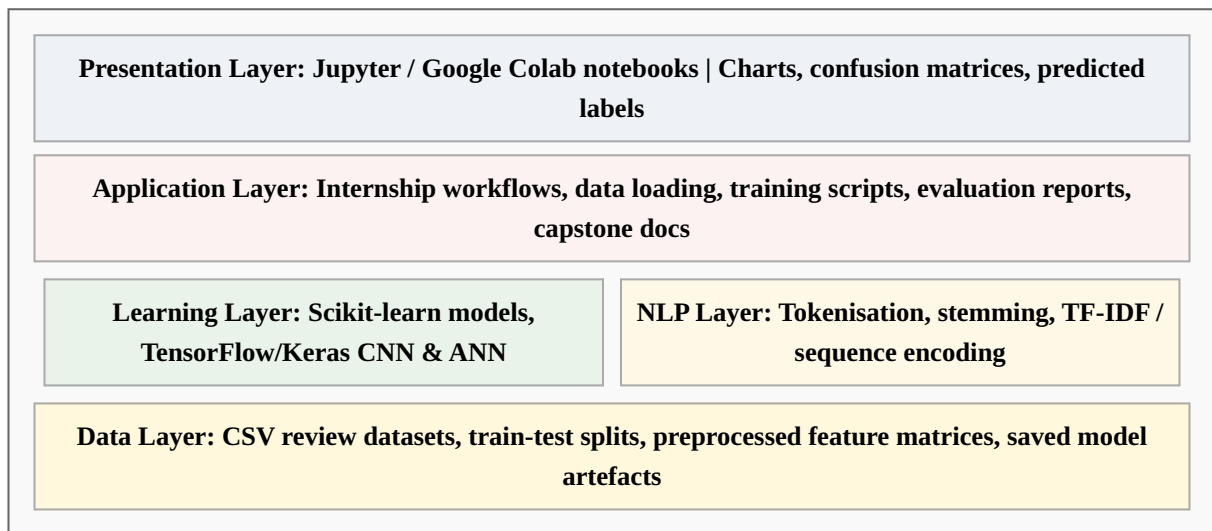


Figure 4.1: System architecture of the sentiment analysis pipeline

Information flow is unidirectional during training: raw text is normalised, converted to TF-IDF or sequence tensors, passed to an estimator, and scored on a held-out test set. At inference time the fitted vectoriser and model are reused so that a new review follows the same path.

4.8 Methodology

The methodology is both chronological (four academic weeks matching the host syllabus) and technical (a standard CRISP-style sequence of understanding, preparation, modelling and evaluation). Dates below follow the certificate window 08/08/2026–08/09/2026.

Week	Dates	Syllabus Focus	Internship Activity
Week 1	08–14 Aug 2026	Foundations of AI; Python; NumPy; Pandas; cleaning; visualisation	Environment set-up; EDA on tabular and text-ready datasets; documentation of data quality issues
Week 2	15–21 Aug 2026	Supervised learning; linear & logistic regression; evaluation & optimisation	Baseline predictors; train-test split; accuracy, precision, recall, F1; cross-validation notes
Week 3	22–28 Aug 2026	Neural networks; TensorFlow/Keras; CNN for image data	Shallow ANN experiments; baseline CNN for image classification to contrast with text models
Week 4	29 Aug–08 Sep 2026	NLP; sentiment analysis; introduction to RL/Q-learning; capstone	Review-text pipeline; sentiment classifier; capstone write-up and certificate closure

Table 4.1: Weekly work plan mapped to the approved AI internship syllabus

4.8.1 Week 1 — Foundations, Python and data preparation: Week 1 established vocabulary: narrow versus general AI, the distinction between rule-based systems and learners, and typical industry applications. Laboratory

work concentrated on Python data structures, NumPy arrays and Pandas DataFrames. Data cleaning drills included missing-value strategies, type coercion, duplicate removal and normalisation. Visualisation with Matplotlib and Seaborn was used to inspect class imbalance before fitting models.

4.8.2 Week 2 — Classical machine learning and evaluation: Week 2 implemented linear regression for continuous targets and logistic regression for binary labels with scikit-learn. Emphasis was placed on train-test splitting, underfitting/overfitting, and evaluation metrics.

Metric	Applicable Task	Interpretation Used in Internship
Accuracy	Classification	Overall share of correctly labelled instances
Precision	Classification	Reliability of positive (e.g. positive-sentiment) predictions
Recall	Classification	Coverage of actual positive class instances
F1-score	Classification	Harmonic mean used when class balance is imperfect
MAE / RMSE	Regression	Average and squared error for continuous targets
Confusion Matrix	Classification	Error-type diagnosis for sentiment and image experiments

Table 4.3: Evaluation metrics adopted for supervised learning experiments

4.8.3 Week 3 — Neural networks and convolutional models: Week 3 introduced perceptrons, hidden layers, activation functions, loss-driven training and backpropagation. A compact feed-forward network was built in Keras, along with a baseline Convolutional Neural Network (CNN) for image classification to contrast spatial learning with text processing.

4.8.4 Week 4 — NLP, reinforcement learning orientation and capstone: Week 4 focused on language processing. Reviews were lower-cased, tokenised, and processed with TF-IDF vectorisation before being classified using logistic regression and dense neural networks. A short Q-learning exercise illustrated reinforcement learning principles.

Learning Module	Capstone Artefact
Python, Pandas, Visualisation	Cleaned review table, class-balance plots, missing-value report
Supervised ML (scikit-learn)	Logistic regression / linear model baseline for polarity prediction
Model Evaluation	Hold-out test metrics, cross-validation summary, confusion matrix
Neural Networks (Keras)	Optional dense network on vectorised text for comparison
CNN (Image Track)	Separate image-classifier notebook demonstrating spatial feature learning
NLP (NLTK/spaCy, TF-IDF)	Token pipeline and feature matrix feeding the sentiment model
Reinforcement Learning Intro	Short Q-learning notebook (conceptual; not used in production)

Table 4.4: Capstone mapping: learning modules to project artefacts

4.9 Expected Outcomes

- A reproducible notebook loading reviews, applying preprocessing, and training linear and neural classifiers.
- A held-out evaluation protocol reporting accuracy, precision, recall, F1-score and confusion matrix.
- Demonstrable coverage of the host syllabus across Python, ML, ANN, CNN, NLP and RL.
- Visual artefacts (class-balance plots, metric tables, sample predictions) suitable for viva voce explanation.
- Clear statement of limitations: English-only scope, sensitivity to sarcasm, and domain shift.
- Formal completion of the AICTE and ICAC approved internship, evidenced by the certificate in Chapter 3.

4.10 Certificates of Completion and Verification Particulars

Particular	Recorded Value
Intern	DISHA CHAUDHARY (Roll No.: 25SCS1003001639)
Organisation	Codec Technologies Pvt. Ltd.
Designation	Artificial Intelligence Intern
Duration	08/08/2026 to 08/09/2026 (One Month)
Approval	AICTE and ICAC approved internship programme
Portal	National Internship Portal (Ministry of Education)
Programme Manager	Dr. Anurag Shrivastava
AICTE ID	CORPORATE6759d549ce50e1733940553
Offer Letter NCS ID	E19E86-0116588288923 (Issued: 08/08/2026)
Partner Mark	Google for Education Partner ISO 9001:2015 Certified

Table 4.5: Internship verification particulars from completion certificate and offer letter

OFFER LETTER ATTACHMENT

Official Internship Offer Letter issued by Codec Technologies Pvt. Ltd.:

CHAPTER 5

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